MING YIN

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EDUCATION

Duke University — *Aug 2024 – Present*
Ph.D. in Electrical and Computer Engineering · Advisor: Prof. Yiran Chen — Durham, NC

University of Science and Technology of China (USTC) — *Sept 2020 – June 2024*
B.E. in Computer Science, School of the Gifted Young · Advisor: Prof. Weiming Zhang — Hefei, China

RESEARCH EXPERIENCE

Meta, Research Scientist Intern — *May 2026 – Dec 2026*
LLM post-training for recommendation; long-horizon LLM agents — Menlo Park, CA

- Post-training LLMs for instruction-guided generative recommendation. First-authored *When History Misleads* [6]: asymmetric margin supervision that keeps the model from being misled by a user's history when it conflicts with their explicit instruction.
- Agentic RL for long-horizon LLM agents: co-authored *Self-Credit* [5], which turns the agent's own context-compaction summaries, with citations to earlier turns, into turn-level credit, rescaled by comparing successful and failed rollouts within each group. Beats the strongest agentic RL baselines by 8.5 and 5.6 points on BrowseComp-Plus (Qwen3-8B / 32B).
- Post-training long-horizon LLM agents (ongoing).

Zoom, Research Intern — *May 2025 – Aug 2025*
Agent tool calling & Model Context Protocol (MCP) — Seattle, WA

- Led **LiveMCP-101** [2] (first author), a benchmark of 101 real-world queries that require coordinated, multi-step use of MCP tools, spanning 41 MCP servers and 260 tools (reference plans average 5.4 tool calls, up to 15).
- Designed a parallel evaluation framework: a reference agent executes a validated plan alongside the agent under test, producing real-time reference outputs that stay valid as live tool responses change.
- Benchmarked 18 LLMs. Error analysis identified seven failure modes spanning planning, parameterization and output handling.

PUBLICATIONS

** equal contribution, † corresponding author*

LLM Agents: Agentic RL, Tool Use & Evaluation

[1] **Which Agent Causes Task Failures and When? On Automated Failure Attribution of LLM Multi-Agent Systems**
Shaokun Zhang*, **Ming Yin***, Jieyu Zhang, . . . , Chi Wang, Huazheng Wang, Yiran Chen, Qingyun Wu
ICML 2025 Spotlight

[2] **LiveMCP-101: Stress Testing and Diagnosing MCP-enabled Agents on Challenging Queries**
Ming Yin, Dinghan Shen, Silei Xu, Sixun Dong, . . . , Kaiqiang Song†, Yiran Chen† — preprint 2026

[3] **LogicIF: Towards Complex Logic Instruction Following**
Mian Zhang, Shujian Liu, Sixun Dong, **Ming Yin**, . . . , Zhiyu Zoey Chen, Kaiqiang Song — COLM 2026

[4] **Agents' Last Exam**
Yiyou Sun, Xinyang Han, Weichen Zhang, . . . , **Ming Yin**, . . . , Dawn Song — **NeurIPS 2026**

[5] **Self-Credit: Credit Assignment for Long-Horizon Agentic RL via Intra-Group Rescaling**
Fangcong Yin, **Ming Yin**, Shlok Kumar Mishra, . . . , Karthik Abinav Sankararaman, Greg Durrett, Dinkar Juyal
under review

Model training & Recommendation

[6] **When History Misleads: Asymmetric Margin Supervision for Instruction-Guided LLM Generative Recommendation**
Ming Yin, Yuhan Yang, Chen Chen, Xinyu Lin, Wentao Shi, Fangcong Yin, . . . , Yiran Chen, Qifan Wang
preprint 2026 · work done at Meta

[7] **MARS: Multi-resolution Adaptive Routing for Sequential Recommendation**
Ming Yin, Sixun Dong, Yudong Liu, Wen-Yun Yang, Yunjiang Jiang, Yiran Chen — **NeurIPS 2026**

Efficient Multimodal LLMs

[8] **MMTok: Multimodal Coverage Maximization for Efficient Inference of VLMs**
Sixun Dong, Juhua Hu, Mian Zhang, **Ming Yin**, Yanjie Fu, Qi Qian — ICLR 2026

[9] **Keyframe-oriented Vision Token Pruning: Enhancing Efficiency of Large Vision Language Models on Long-Form Video Processing**
Yudong Liu, Jingwei Sun, Yueqian Lin, Jingyang Zhang, **Ming Yin**, ..., Hai Li, Yiran Chen — ICCV 2025

Trustworthy ML

[10] **Poisoning Federated Recommender Systems with Fake Users**
Ming Yin, Yichang Xu, Minghong Fang, Neil Gong — WWW 2024

[11] **Robust Federated Learning Mitigates Client-side Training Data Distribution Inference Attacks**
Yichang Xu*, **Ming Yin***, Minghong Fang, Neil Gong — WWW 2024

[12] **LoBAM: LoRA-Based Backdoor Attack on Model Merging**
Ming Yin, Jingyang Zhang, Jingwei Sun, Minghong Fang, Hai Li, Yiran Chen — ICLR 2025 Workshop

[13] **A Few Neurons Reveal When LLMs Misuse Tools: Sparse Detection and Selective Steering for Reliable Tool Use**
Yutong Ke*, **Ming Yin***, Chongwen Zhao, Kaizhu Huang — preprint 2026

Survey

[14] **Seeing, Maintaining, and Learning: The Evolution of Multimodal Agent Memory**
Minghao Guo, Qingyue Jiao, Xiujin Liu, ..., **Ming Yin**, ..., Mengdi Wang, Jiawei Han, Philip S. Yu, Ziyi Ye
under review

HONORS & AWARDS

- **Qiangwei Progress Scholarship** *(Top 1%)*, USTC — *2023*
- **Excellent Student Scholarship, Gold** *(Top 3%)*, USTC — *2021, 2023*
- **ICPC Asia Regional Contest** *(Top 5%)* — *2022*
- **First Prize, Chinese Physics Olympiad (Jiangsu Province)** *(Top 0.04%)* — *2019*

SKILLS

Programming: Python, C/C++, SQL, Bash
ML & LLM tooling: PyTorch, CUDA, MCP (Model Context Protocol), HPC clusters, Linux
Methods: LLM post-training, reinforcement learning, multi-agent systems, robust and federated learning

SERVICE & TEACHING

Reviewer: NeurIPS, ICML, ICLR (2024–present); IEEE TDSC
Teaching Assistant: ECE 661 Computer Engineering ML and Deep Neural Nets, Duke — *Fall 2025*